

YOHANNES NIGUSSE

Product Software Engineer | Software Engineer Intern

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EDUCATION

St. Cloud State University | St. Cloud, MN

B.S. Computer Science, Artificial Intelligence & Machine Learning Track | Expected December 2027

GPA: 3.92 | Coursework: Software Design & Architecture, Object-Oriented Software Development, Relational Databases, Data Structures & Algorithms

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Java, C#, Bash, HTML, CSS

Full Stack & APIs: React, FastAPI, Flask (coursework), REST APIs and REST API design (coursework), Node.js, Django, web applications, backend services, web services, microservices, Streamlit, PySide6/QML

Data & AI/ML: PostgreSQL, SQLite, SQLAlchemy ORM, MongoDB, Redis, schema design, data modeling, pgvector, vector search, data structures, LLMs, RAG, agents, algorithms, AI APIs (OpenAI/Gemini)

Cloud & Delivery: Docker, Git, GitHub Actions, CI/CD, AWS, Azure, cloud services (AWS Lambda/API Gateway, Azure Storage/Search, GCP), Linux, GCP

Testing & Quality: Unit testing, automated testing, integration testing, regression testing, debugging, root-cause analysis, QA, validation, code review, documentation, accessibility, OOXML

Development & Operations: AI-assisted development (Claude Code), Agile work tracking, collaboration, technical communication, operational monitoring, logs, telemetry, reliability

PROFESSIONAL EXPERIENCE

Accenture | Seattle, WA | **Technology Summer Analyst | June 2026 - August 2026**

- Resolved 10+ externally reported accessibility defects in TypeScript/OOXML production software backed by Azure storage/search, unblocking a release milestone for approximately 30,000 internal accounts.
- Validated accessibility fixes across 8 formal scenarios using automated regression testing and NVDA — covering reading order, alt text, title structure, navigation, footnotes, and malformed bullets — producing sign-off evidence that cleared the production release gate.
- Extended the pricing tool's production API integration layer between systems to carry additional data inputs into structured recommendation documents; migrated document generation to AWS Lambda/API Gateway.

Kibur College | Remote | **Software Engineering Intern | January 2026 - May 2026**

- Built and owned an end-to-end Telegram/email RAG assistant combining semantic search over ML embeddings in PostgreSQL/pgvector with Gemini LLM integration; validated across 84 queries from 12 users in an owner-confirmed pilot.
- Benchmarked retrieval and response accuracy across 75 questions: 68/75 retrieved expected top-five context and 70/75 answers were correct or partially correct (owner-confirmed).
- Designed relational schema, pgvector indexing, and retrieval routing to isolate institutional facts from generated responses; built a SQLite admin dashboard enabling approximately 12 staff to update the knowledge base without code changes.

Brain-Computer Interface Lab, St. Cloud State University | St. Cloud, MN | **Undergraduate Research Assistant | August 2024 - May 2025**

- Identified the root cause of EEG sample-rate mismatches and implemented metadata-driven polyphase resampling, restoring 14 previously failed recordings across 3 runs; all 14 produced valid 8-channel PSD features for the anonymized dataset.
- Wrote RSA key-management scripts that automated key expiry and blocked post-login authorized_keys changes across two workstations and one storage server, hardening HPC access controls for four researchers.
- Debugged and corrected tab resizing, active-state indicators, and button alignment across an eight-tab PySide6/QML research interface, restoring consistent drone, robot, and model controls.

Center for Women in Technology, UMBC | Baltimore, MD | **Technical Operations Assistant | September 2023 - May 2024**

- Automated attendance tracking for 50-100 participants per session with Python and the Google Sheets API, reducing manual record processing by approximately 85%.

PROJECTS

CampusMarket | **Lead Developer, 5-person course team | Python, FastAPI, React, PostgreSQL, Claude Code**

- Designed and led full-stack architecture for a campus marketplace's listing/search workflows across React, FastAPI, and PostgreSQL, applying SOLID and Factory, Observer, and Strategy patterns; the structure enabled separation of concerns across the five-person team.
- Leveraged Claude Code to scaffold listing/search CRUD endpoints, accelerating initial delivery by 40%; conducted 20+ pull-request reviews focused on business requirements, edge cases, and code quality.

The Stack | **Solo Developer | Python, PostgreSQL, OpenAI API, Docker, Streamlit**

- Built an end-to-end AI data pipeline ingesting YouTube and event feeds, ranking summaries with an OpenAI curator agent, storing results in PostgreSQL, and delivering Gmail digests via Streamlit; implemented item-level error isolation so a single failure did not halt a run.
- Executed load and reliability testing across 500 runs on a reproducible input set, achieving 96.4% successful runs; built operational monitoring commands and structured JSON logging to trace degraded/failure cases to API rate limits.

NimbusQueue | **Backend Lead, 4-person team | CSCI 312: Fundamentals of Distributed Systems | Python, FastAPI, PostgreSQL, Redis, Docker, GCP, GitHub Actions**

- Built a FastAPI service interface and PostgreSQL-backed task-queue microservice for asynchronous background-job processing across three GCP workers, sustaining load-tested bursts of 1,000+ queued jobs.
- Automated integration testing and deployment on merges to main with Docker Compose and GitHub Actions, removing manual deployment steps for the four-person team.
- Benchmarked scalability on 5,000 synthetic jobs: three workers reduced queue-drain time by 62% versus one worker; compared worker-processing P95 latency to guide performance optimization.

InvenTrack | **Solo Developer | CSCI 331: Software Systems | Python, SQLAlchemy, SQLite, Streamlit**

- Built a Streamlit inventory app with SQLAlchemy ORM and SQLite for stock, reorder points, and transaction tracking; decoupled domain logic from storage to enable in-memory testing, where unit and integration tests caught two reorder-point data-integrity bugs before the course demo.

LEADERSHIP & HONORS

Vice President, National Society of Black Engineers | ColorStack Affiliate | Student Government Technology Fee Committee | Dean's List | Monte Johnson CS Scholarship | Duane Joyer Memorial Scholarship